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$$T_4\left(f, \frac{\pi}{2}\right)(x) \text{ voor } f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f'''(x) = -\cos x$$

$$f''''(x) = \sin x$$

$$T_4\left(f, \frac{\pi}{2}\right)(x) = \sum_{k=0}^4 \frac{f^{(k)}\left(\frac{\pi}{2}\right)}{k!} \left(x - \frac{\pi}{2}\right)^k$$
$$1 - \frac{1}{2} \left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24} \left(x - \frac{\pi}{2}\right)^4$$

References